

<https://practicetestautomation.com/practice-test-login/> login page

<https://www.saucedemo.com/> dummy website for testing(login)

SOFTWARE TESTING (ST / STQA) PRACTICAL FILE

PRACTICAL 3

Basic Selenium program to launch browser and open a website

◆ AIM

To write and test a basic Selenium program to launch a browser and open a web application.

◆ PROGRAM (PRACTICAL-3)

```
package demo1;
```

```
import org.openqa.selenium.WebDriver;  
import org.openqa.selenium.firefox.FirefoxDriver;  
import org.openqa.selenium.firefox.FirefoxOptions;
```

```
public class demo1 {
```

```
    static String driverPath =  
        "D:\\Selenium\\geckodriver-v0.34.0-win32\\geckodriver.exe";
```

```
    public static void main(String[] args) {
```

```
        int a = 10, b = 20;  
        System.out.println("HI");  
        System.out.println(a + b);  
        System.out.println("Selenium demo.....");
```

```
        // Set GeckoDriver path  
        System.setProperty("webdriver.gecko.driver", driverPath);
```

```
        // Firefox Options  
        FirefoxOptions options = new FirefoxOptions();
```

```
options.setCapability("marionette", true);

// Launch Firefox
WebDriver driver = new FirefoxDriver(options);

driver.get("https://www.facebook.com/");
driver.manage().window().maximize();

driver.quit();
}
}
```

♦ OUTPUT

Browser launches and Facebook home page is displayed.

♦ CONCLUSION

Thus, the Selenium program successfully launches the browser and opens the web page.

PRACTICAL 4

Write and test a program to perform login operation and validate the result

♦ AIM

To write and test a Selenium program to perform login operation and validate whether the login is successful.

♦ SOFTWARE REQUIREMENTS

- Java
- Selenium WebDriver
- Mozilla Firefox
- GeckoDriver

◆ PROGRAM (PRACTICAL-4)

```
package demo;

import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.firefox.FirefoxDriver;

public class Practical4_LoginValidation {

    public static void main(String[] args) {

        // Set GeckoDriver path
        System.setProperty(
            "webdriver.gecko.driver",
            "C:\\Users\\Student\\Downloads\\geckodriver.exe"
        );

        // Launch Firefox browser
        WebDriver driver = new FirefoxDriver();

        // Open login page
        driver.get("https://practicetestautomation.com/practice-test-login/");

        // Enter username
        driver.findElement(By.id("username"))
            .sendKeys("student");

        // Enter password
        driver.findElement(By.id("password"))
            .sendKeys("Password123");

        // Click login button
        driver.findElement(By.id("submit"))
            .click();

        // Get page title
        String title = driver.getTitle();
        System.out.println("Page Title: " + title);

        // Validate login
        if (title.contains("Logged In Successfully")) {
            System.out.println("LOGIN SUCCESSFUL");
        } else {
```

```
        System.out.println("LOGIN FAILED");
    }

    driver.quit();
}
}
```

◆ OUTPUT

Page Title: Logged In Successfully | Practice Test Automation
LOGIN SUCCESSFUL

◆ CONCLUSION

Thus, the program successfully performs login operation and validates the result.

✓ PRACTICAL 5

Update Student Records in Excel (Pass/Fail) using TestNG and JXL

◆ PROGRAM (PRACTICAL-5)

```
import java.io.FileInputStream;
import java.io.FileOutputStream;

import jxl.Workbook;
import jxl.Sheet;
import jxl.Cell;

import jxl.write.Label;
import jxl.write.WritableSheet;
import jxl.write.WritableWorkbook;

import org.testng.annotations.Test;

public class StudentRecord {

    @Test
    public void updateResult() throws Exception {

        FileInputStream fi =
```

```

        new FileInputStream("C:\\Student.xls");
Workbook wb = Workbook.getWorkbook(fi);
Sheet sh = wb.getSheet(0);

FileOutputStream fo =
    new FileOutputStream("C:\\Result.xls");
WritableWorkbook ww = Workbook.createWorkbook(fo);
WritableSheet ws = ww.createSheet("Result", 0);

for (int i = 0; i < sh.getRows(); i++) {
    for (int j = 0; j < sh.getColumns(); j++) {
        Cell c = sh.getCell(j, i);
        ws.addCell(new Label(j, i, c.getContents()));
    }
}

ws.addCell(new Label(6, 0, "Result"));

for (int i = 1; i < sh.getRows(); i++) {
    int marks =
        Integer.parseInt(sh.getCell(4, i).getContents());

    if (marks > 35)
        ws.addCell(new Label(6, i, "Pass"));
    else
        ws.addCell(new Label(6, i, "Fail"));
}

ww.write();
ww.close();

System.out.println("Records updated");
}
}

```

PRACTICAL 6

Select students who scored more than 60 in any subject

PROGRAM (PRACTICAL-6)

```

import jxl.write.Label;
import jxl.write.WritableSheet;

```

```

import jxl.write.WritableWorkbook;
import java.io.*;
import jxl.*;
import org.testng.annotations.*;

public class Student_Record {

    @Test
    public void testImportexport1() throws Exception {
        FileInputStream fi = new FileInputStream("C:\\Users\\Admin\\Desktop\\Students1.xls");
        Workbook w = Workbook.getWorkbook (fi);
        Sheet s=w.getSheet(0);

        FileOutputStream fo = new FileOutputStream("C:\\Users\\Admin\\Desktop\\Result1.xls");
        WritableWorkbook ww = Workbook.createWorkbook (fo);
        WritableSheet ws = ww.createSheet ("result1", 0);
        int newRow = 0;

        for (int i = 1; i < s.getRows(); i++) {
            int cMarks = Integer.parseInt(s.getCell(2, i).getContents());
            int dMarks = Integer.parseInt(s.getCell(3, i).getContents());
            int eMarks = Integer.parseInt(s.getCell(4, i).getContents());

            if (cMarks >= 60 || dMarks >= 60 || eMarks >= 60) {
                for (int j = 0; j < s.getColumns(); j++) {
                    String value = s.getCell(j, i).getContents();
                    ws.addCell(new Label(j, newRow, value));
                }
                newRow++;
            }
        }
        ww.write();
        ww.close();
    }
}

```

PRACTICAL 7

Count total number of objects present on a web page

◆ PROGRAM (PRACTICAL-7)

package demo;

```

import org.openqa.selenium.WebDriver;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.firefox.FirefoxOptions;
import org.openqa.selenium.By;
import org.openqa.selenium.WebElement;
import java.util.List;

public class Practical7_ObjectCount {

    static String driverPath =
        "C:\\Users\\student\\Downloads\\geckodriver.exe";

    public static void main(String[] args) {

        System.setProperty("webdriver.gecko.driver", driverPath);

        FirefoxOptions options = new FirefoxOptions();
        WebDriver driver = new FirefoxDriver(options);

        driver.get("https://www.google.com/");

        List<WebElement> links =
            driver.findElements(By.xpath("//a"));

        List<WebElement> buttons =
            driver.findElements(By.xpath("//input[@type='submit']"));

        System.out.println("Number of links = " + links.size());
        System.out.println("Number of buttons = " + buttons.size());

        driver.quit();
    }
}

```

PRACTICAL 8

Get number of items in a list / combo box

PROGRAM (PRACTICAL-8)

```

package demo;

import java.util.List;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;

```

```

import org.openqa.selenium.WebElement;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.support.ui.Select;

public class Practical8_ListCount {

    public static void main(String[] args) {

        System.setProperty(
            "webdriver.gecko.driver",
            "C:\\Users\\student\\Downloads\\geckodriver.exe"
        );

        WebDriver driver = new FirefoxDriver();

        driver.get("https://www.facebook.com/campaign/landing.php");

        Select se = new Select(
            driver.findElement(By.xpath("//select[@id='month']"))
        );

        List<WebElement> mylist = se.getOptions();

        System.out.println("Number of items = " + mylist.size());

        driver.quit();
    }
}

```

PRACTICAL 9

Count checked and unchecked checkboxes

```

import java.util.List;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;

public class CheckBoxCount {

    public static void main(String[] args) {

```

```
WebDriver driver = new ChromeDriver();

driver.get("https://the-internet.herokuapp.com/checkboxes");

List<WebElement> checkboxes =
    driver.findElements(By.xpath("//input[@type='checkbox']"));

int checked = 0;
int unchecked = 0;

for (WebElement box : checkboxes) {
    if (box.isSelected())
        checked++;
    else
        unchecked++;
}

System.out.println("Checked = " + checked);
System.out.println("Unchecked = " + unchecked);

driver.quit();
}
}
```